# MOHAND ATTYA

## Systems & Biomedical Engineer | AI & Medical Imaging

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Mohand Emad | Github

### EDUCATION:

**Bachelor's Degree in Systems and Biomedical Engineering,** *Cairo University* 2020 – 2025

- **GPA:** 3.5/4 **Specialization:** AI

### PROJECTS:

**TherAInostics: Web-Based Deep Learning System for Evaluating Lu-PSMA Therapy Efficacy in Prostate Cancer Using PET/CT** , *Graduation Project*

Developed and evaluated 3 state-of-the-art deep learning pipelines for PET/CT medical image analysis to automate and enhance physician tracking in prostate cancer:

- **Attention UNet** for prostate segmentation (**0.85 Dice**)
- **3D UNet** for lesion segmentation (**0.67 Dice**)
- **ResNet50** for lesion classification (**73% accuracy**)

Integrated all models into an intuitive DICOM viewer, enabling seamless visualization and workflow integration with PACS in hospitals and clinics.

**Tools:** MONAI, Python, PyTorch, OHIF, Orthanc

**Face Recognition Attendance Web System** , *IoT Project*

Developed an IoT system using ESP32-CAM and ESP8266, integrated with a web app, to automatically track lecture hall attendance via facial recognition.

**Tools:** C, HTML, JavaScript, FastAPI, ArcFace, OpenCV

**SanOris: Dental Management System** , *HIS Project*

Implemented Authentication Tools for sign-in and sign-out pages.

**Tools:** Vite React, TypeScript, Node.js, Express.js, MySQL, Docker, Zustand, Prisma, Firebase

**FourierBlend Studio: Desktop-Based Image Mixer** , *DSP Project*

**Tools:** FFT, Image Processing, Pillow, PyQt,

### PROFESSIONAL EXPERIENCE:

**Biomedical & Systems Engineer,** *Al Galaa Military Medical Complex* 11/2025 – 12/2026

- Manage operational integrity, troubleshooting, and maintenance for a large inventory of critical medical devices across high-stress hospital departments, including ICUs, Radiology, and Oncology.
- Collaborate closely with clinical staff, physicians, and external technical support teams to resolve critical hardware and workflow bottlenecks

**Biomedical Engineer Intern,** *Cleopatra Hospitals Group* 07/2024 – 09/2024

- Contributed to the digital transformation and digitalization of internal communication workflows and hardware inventory systems across hospital branches

**Project Manager,** *Enactus Cairo* 09/2021 – 09/2022

- *Lead the projects team to establish **GellyFish** Project—Extracting Gelatin and Burning Bandages from fish skin in collaboration with the Faculty of Veteran Medicine, CU.* (*2nd. Place holder in the National Competition* 2022)

### TECHNICAL SKILLS:

**Systems & Software:**
Python - TypeScript - SQL - Bash - React - Node.js - Flask Express - FastAPI - Docker - MySQL - Firebase

**AI & Computer Vision:**
PyTorch - MONAI - OpenCV - UNet - Attention - UNet - ResNet - YOLO - SAM - MedSAM - Keras

**Medical Imaging**
PET/CT - DICOM - Medical Image Segmentation - OHIF - Orthanc - PACS

**Embedded Systems:**
C/C++ - ARM - ESP 8266 & 32 - Embedded Systems - CMake

### Certificates

**IELTS Academic Certification**

- Overall Band Score: 7.5

**LangChain Certified Agent Engineer**